

General Relativity Exercise 6

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1 Warmup

Starting from the Einstein equation

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = T_{\mu\nu} \quad (1)$$

Taking the trace

$$R - \frac{1}{2}Rg = T \quad (2)$$

Solving for R :

$$R = \frac{2}{1-g}T \quad (3)$$

Assigning this back into the original Einstein's equation and solving for $R_{\mu\nu}$ we get

$$R_{\mu\nu} = T_{\mu\nu} + \frac{g_{\mu\nu}}{1-g}T \quad (4)$$